

MALC Package Manual

Density Estimation from Ordered Log-Concave Discrete Data

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1 Overview

MALC implements density estimation from observed frequencies on a known grid. The package is designed for application settings where the raw sample is already aggregated into bin counts. The main estimator is `MALC()`, and the package also includes the three comparison methods used in the simulation study:

1. `BK2002()`
2. `BinnedNP()`
3. `KernSmooth()`

All fitting functions use the same basic input structure:

- `counts`: observed frequencies for each grid interval.
- `grid`: numeric grid breaks with length `length(counts) + 1`.

The MALC fit supports density, distribution, quantile, mean, and plotting methods. Kernel comparison fits support L2 evaluation through `Eval()` and visualization through `Plot.Kernel()`.

2 Installation

The package can be installed from GitHub while the package branch is under development:

```
install.packages("remotes")
remotes::install_github("FurkanDanisman/MALC")
```

Then load the package:

```
library(MALC)
```

3 Input Format

`MALC()` does not accept raw observations. It accepts only observed frequencies and grid breaks.

```
set.seed(1)
x <- rnorm(1000)
grid <- seq(-4, 4, by = 0.5)
counts <- as.integer(table(cut(x, breaks = grid, include.lowest = TRUE)))
```

In direct package usage, the preprocessing step above can be done by the user or by an application before calling the package.

4 MALC Estimator

4.1 Fit

```
fit <- MALC(counts, grid)
fit_s <- MALC(counts, grid, smooth = TRUE)
```

Usage

```
MALC(counts, grid, smooth = FALSE, B = 10000, alpha = 2,  
      seed = 20180621, log_conc = TRUE)
```

Arguments

Argument	Description
counts	Integer observed frequencies for each bin.
grid	Numeric bin breaks. Must have length <code>length(counts) + 1</code> .
smooth	Logical. If <code>TRUE</code> , downstream density functions use the smoothed fit.
B	Number of Monte Carlo draws used internally.
alpha	Shape parameter used in the beta perturbation.
seed	Random seed used internally.
log_conc	Passed to <code>logcondens::logConDens()</code> .

5 MALC Methods

5.1 dmalc()

Evaluate the fitted MALC density.

```
x_eval <- seq(min(grid), max(grid), length.out = 200)  
d_hat <- dmalc(fit, x_eval)
```

Usage

```
dmalc(object, x)
```

5.2 pmalc()

Evaluate the fitted distribution function numerically.

```
pmalc(fit, c(-1, 0, 1))
```

Usage

```
pmalc(object, q, length_grid = 1001)
```

5.3 qmalc()

Evaluate fitted quantiles numerically.

```
qmalc(fit, c(0.25, 0.50, 0.75))
```

Usage

```
qmalc(object, p, length_grid = 1001)
```

5.4 mean_malc()

Return the EM-step mean estimate stored in the fitted object. This is not a smoothed/non-smoothed choice; it is the mean estimate from the MALC fitting step.

```
mean_malc(fit)
```

Usage

```
mean_malc(object)
```

5.5 Plot()

Plot a fitted MALC density. The plotted density follows the `smooth` choice stored in the fitted object.

```
Plot(fit)
Plot(fit_s, col = "red")
```

Usage

```
Plot(object, ...)
plot(x, eval_grid = NULL, add = FALSE, xlab = "x",
     ylab = "Density", col = "navy", lwd = 2, ...)
```

6 Comparison Methods

6.1 BK2002()

Fit the BK2002 comparison estimator from observed frequencies and grid breaks.

```
bk <- BK2002(counts, grid)
```

Usage

```
BK2002(counts, grid, h = NULL, bw_B = 50, bw_seed = 1,
       bw_method = c("ste", "dpi"), max_iter = 200,
       tol_L1 = 1e-6, verbose = FALSE)
```

6.2 BinnedNP()

Fit the BinnedNP comparison estimator. The bandwidth choice is set at fitting time and stored in the fitted object.

```
bn <- BinnedNP(counts, grid, bandwidth = "boot")
bn_plugin <- BinnedNP(counts, grid, bandwidth = "plugin")
```

Usage

```
BinnedNP(counts, grid, bandwidth = c("boot", "plugin"),
  seed_start = 1L, max_tries = 200L)
```

6.3 KernSmooth()

Fit the KernSmooth comparison estimator from observed frequencies and grid breaks.

```
ks <- KernSmooth(counts, grid)
```

Usage

```
KernSmooth(counts, grid, seed = 1, scalest = "minim",
  level = 2L, kernel = "normal", canonical = FALSE,
  truncate = TRUE)
```

7 Kernel Plotting

Plot_Kernel() plots any fitted comparison-method object returned by BK2002(), BinnedNP(), or KernSmooth().

```
Plot_Kernel(bk)
Plot_Kernel(bn, add = TRUE, col = "red")
Plot_Kernel(ks, add = TRUE, col = "blue")
```

Usage

```
Plot_Kernel(object, eval_grid = NULL, add = FALSE,
  xlab = "x", ylab = "Density", col = "darkgreen",
  lwd = 2, ...)
```

8 L2 Evaluation

Eval() computes the numerical L2 distance between a fitted density estimator and a supplied true density function. It accepts both MALC fits and kernel-method fits.

```

true_pdf <- function(z) dnorm(z)

Eval(fit, true_pdf)
Eval(fit_s, true_pdf)
Eval(bk, true_pdf)
Eval(bn, true_pdf)
Eval(ks, true_pdf)

```

Usage

```
Eval(object, pdf, length_grid = 1001)
```

Arguments

Argument	Description
object	Fitted object from MALC(), BK2002(), BinnedNP(), or KernSmooth().
pdf	True density function used for comparison.
length_grid	Number of numerical evaluation points.

9 Complete Example

```

library(MALC)

set.seed(1)
x <- rnorm(1000)
grid <- seq(-4, 4, by = 0.5)
counts <- as.integer(table(cut(x, breaks = grid, include.lowest = TRUE)))

fit <- MALC(counts, grid)
fit_s <- MALC(counts, grid, smooth = TRUE)

dmalc(fit, seq(-2, 2, length.out = 5))
pmalc(fit, c(-1, 0, 1))
qmalc(fit, c(0.25, 0.50, 0.75))
mean_malc(fit)
Plot(fit)

bk <- BK2002(counts, grid)
bn <- BinnedNP(counts, grid, bandwidth = "boot")
ks <- KernSmooth(counts, grid)

Plot_Kernel(bk)
Plot_Kernel(bn, add = TRUE, col = "red")
Plot_Kernel(ks, add = TRUE, col = "blue")

true_pdf <- function(z) dnorm(z)
Eval(fit, true_pdf)
Eval(fit_s, true_pdf)
Eval(bk, true_pdf)

```

```
Eval(bn, true_pdf)
Eval(ks, true_pdf)
```

10 Exported Functions

- `MALC()`
- `dmalc()`
- `pmalc()`
- `qmalc()`
- `mean_malc()`
- `Plot()`
- `BK2002()`
- `BinnedNP()`
- `KernSmooth()`
- `Plot_Kernel()`
- `Eval()`